

CS410 Midterm 3 Part (a)

3 January 2024

I. MULTIPLE CHOICE QUESTIONS (40 POINTS)

1. Which of the following is not correct for Turing machines?

- a) A Turing machine can both write on tape and read from it.
- b) The head can move both to the left and to the right.
- c) The tape is infinite.
- d) A multi-tape Turing machine is more powerful than a single tape Turing machine.
- e) All of them correct.

Given the following Turing machine description, answer questions 2-4 accordingly. M_1 for the language $B = \{a^n b^n c^n | n \geq 0\}$ as:

On input string w ;

- a. Sweep from start to end if symbols are out of order, reject.
- b. Move head to start, search for a, if found cross it else goto step e.
- c. Move head to start, search for b, if found cross it and goto step b, else reject.
- d. Move head to start, search for c, if found cross it and goto step b, else reject.
- e. Move head to start, search for b or c, if found reject, else accept.

2. Which of the following string is possible while iterating the Turing machine on input aaabbbccc?

- a) xaaxbbxxc
- b) xxaxxbxxx
- c) axabxbxc
- d) xxxxbxxc
- e) xxaxxbxcx

3. Which of the following string is on the tape just before the Turing machine accepts on input aabbcc?

- a) xxxxxc
- b) aabbcc
- c) xaxbxc
- d) xxxxxx
- e) xxxbxc

4. Which of the following string is on the tape just before the Turing machine rejects on input abbcc?

- a) xxxxx
- b) xxbxc
- c) abbcc
- d) xxxxc
- e) xbxcx

5. Which of the following string is a possible configuration for a Turing machine?

- a) $10q_11101101$
- b) $1011q_201101$
- c) $q_0101101101$
- d) All of the above
- e) None of the above

6. If we apply the state transition function $\delta(q_i, b) = (q_j, c, R)$ to the configuration $uaq_i bv$, which configuration will we have next?

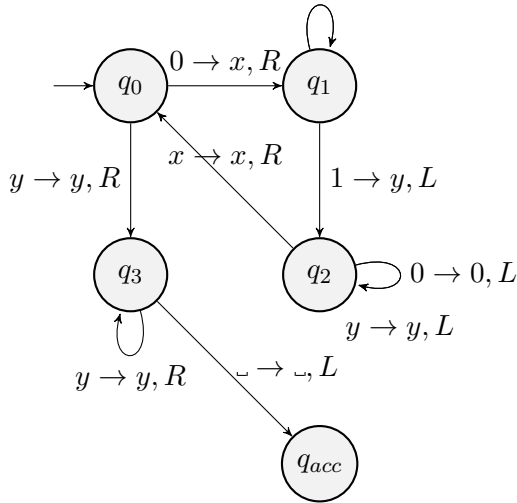
- a) $uq_j acv$
- b) $uaq_j cv$
- c) $uaq_j bcv$
- d) $uacq_j v$
- e) None of the above

7. If we apply the state transition function $\delta(q_i, b) = (q_j, c, L)$ to the configuration $uaq_i bv$, which configuration will we have next?

- a) $uq_j acv$
- b) $uaq_j cv$
- c) $uaq_j bcv$
- d) $uacq_j v$
- e) None of the above

Given the following Turing machine TM_1 . Answer questions 8-11 accordingly.

$y \rightarrow y, R$
 $0 \rightarrow 0, R$



8. What is Q ?

- a) $\{q_0, q_1, q_2, q_3\}$
- b) q_0
- c) $\{q_{acc}, q_{rej}\}$
- d) $\{q_0, q_1, q_2, q_3, q_{acc}, q_{rej}\}$
- e) $\{q_{acc}, q_{rej}\}$

9. What is Σ ?

- a) $\{0, 1\}$
- b) $\{0, 1, x, y\}$
- c) $\{0, 1, x, y, \neg\}$
- d) $\{0, 1, x, y, \neg, L\}$
- e) $\{0, 1, x, y, \neg, L, R\}$

10. What is τ ?

- a) $\{0, 1\}$
- b) $\{0, 1, x, y\}$
- c) $\{0, 1, x, y, \neg\}$
- d) $\{0, 1, x, y, \neg, L\}$
- e) $\{0, 1, x, y, \neg, L, R\}$

11. In how many possible ways can the heads of a k tape multi-tape Turing machine move, if there are three moves available L, R, S?

- a) $k!$
- b) 2^k
- c) 3^k
- d) k^3
- e) k^2

12. Let say we have 2-tape Turing machine with tape contents $[1\ 0\ 0\ 1\ \sqcup]$ and $[c\ b\ a\ \sqcup]$. Let also say that currently the first tape head shows the second 0, and the second tape head shows c. What will the single tape version of this 2-tape Turing machine?

- a) $[1\ 0\ 0\ 1\ c\ b\ a]$
- b) $[1\ 0\ 0\ 1\ \sqcup\ c\ b\ a\ \sqcup]$
- c) $[1\ 0\ 0\ 1\ c\ b\ a\ \sqcup]$
- d) $[1\ 0\ \dot{0}\ 1\ \dot{c}\ b\ a\ \sqcup]$
- e) $[1\ 0\ \dot{0}\ 1\ \# \dot{c}\ b\ a\ \# \sqcup]$

13. Which algorithm do we utilize in simulation the Nondeterministic TM with multi-tape deterministic TM?

- a) shortest path
- b) depth first search
- c) breadth first search
- d) minimum spanning tree
- e) None of the above

14. Which of the following languages is not decidable?

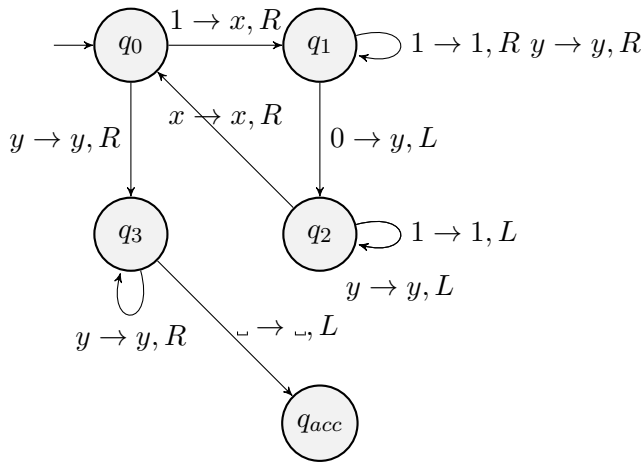
- a) Complement of a decidable language
- b) Union of two decidable languages
- c) Concatenation of two decidable languages
- d) Star of a decidable language
- e) None of the above

15. Which of the following languages is not decidable?

- a) $A = \{ \langle G, A \rangle \mid G \text{ is a DFA and } L(A) = \emptyset \}$
- b) Every regular language
- c) Every Turing-recognizable language
- d) Every context free language
- e) None of the above

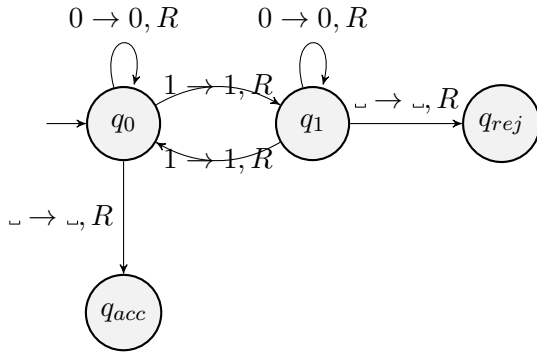
16. Which of the following sets is the largest?

- a) Regular languages
- b) Context-free languages
- c) Decidable languages
- d) Turing-recognizable languages
- e) All of them equal in size



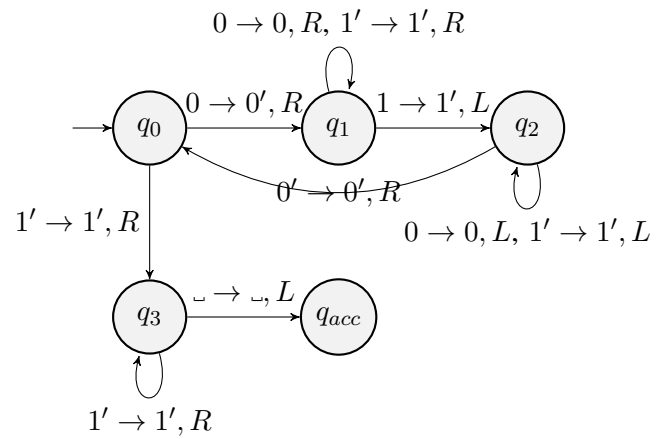
17. Which of the following languages is depicted in the TM given above?

- a) $L = \{0^{2n}1^n \mid n \geq 1\}$
- b) $L = \{0^n1^n \mid n \geq 1\}$
- c) $L = \{0^n1^n0^n \mid n \geq 1\}$
- d) $L = \{1^n0^n \mid n \geq 1\}$
- e) $L = \{1^{2n}0^n \mid n \geq 1\}$



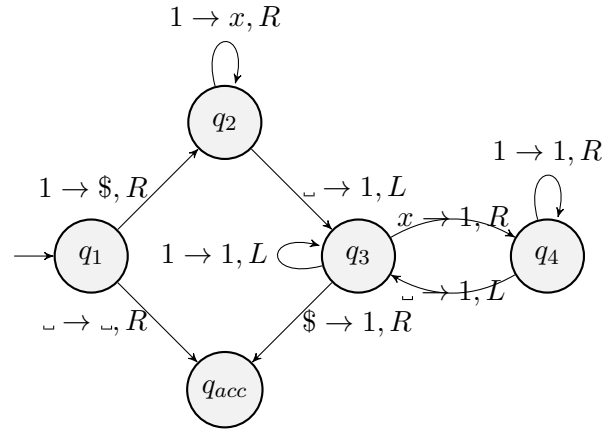
18. Which of the following languages is depicted in the TM given above?

- a) $L = \{w \mid \text{Number of 1's in } w \text{ is a multiple of 3}\}$
- b) $L = \{0^n1^n \mid n \geq 1\}$
- c) Language of palindromes of even length
- d) $L = \{1^n0^n \mid n \geq 1\}$
- e) $L = \{w \mid \text{Number of 1's in } w \text{ is a multiple of 2}\}$



19. Which of the following languages is depicted in the TM given above?

- a) $L = \{w \mid \text{Number of 1's in } w \text{ is a multiple of 3}\}$
- b) $L = \{0^n1^n \mid n \geq 1\}$
- c) Language of palindromes of even length
- d) Given the input as 1^* doubles the number of symbols.
- e) None of the above



20. Which of the following languages is depicted in the TM given above?

- a) $L = \{w \mid \text{Number of 1's in } w \text{ is a multiple of 3}\}$
- b) $L = \{0^n1^n \mid n \geq 1\}$
- c) Language of palindromes of even length
- d) Given the input as 1^* doubles the number of symbols.
- e) None of the above